

KANSER Paneli: Advanced Stacking Raporu

NB30 | 2026-06-22

Yoneticici Ozeti

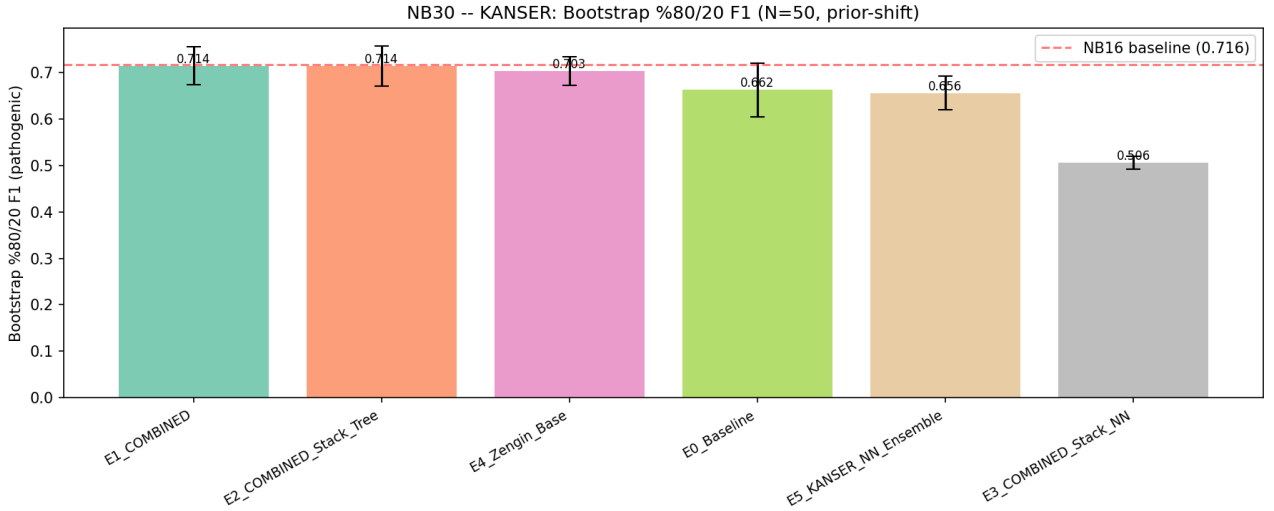
KANSER panelinin (n=385, pos=265, neg=120) %80/20 F1 skorunu yükseltmek için 5 deney (E0-E5) karşılaştırıldı. En iyi: E1_COMBINED (Boot=%80/20 F1=0.7143, AUPRC=0.9643). NB16 referans: 0.716. Fark: -0.0017 -> NB16'ya yakın.

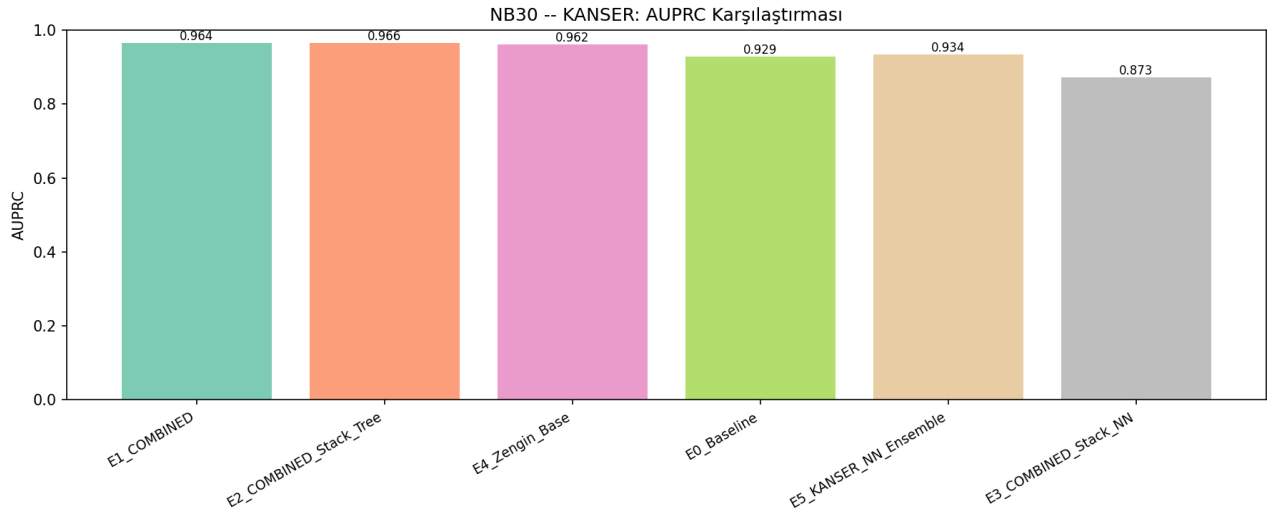
Stacking: COMBINED (MASTER+PAH+CFTR) eğitilip KANSER test. Meta-learner = Logistic Regression (L2). Base modeller: tree (LGBM,XGB,RF,CB) + NN/DNN. Threshold: %80/20 benign-agirlikli bootstrap robust. Prior-shift: Saerens 2002 (pi_test=0.20).

1. Deney Karsilastirmasi (Boot-mean sirali)

Deney	Boot-F1	Boot-std	AUPRC	Prec	Recall	FP	FN
E1_COMBINED	0.714	0.041	0.964	0.943	0.804	13	52
E2_COMBINED_Stack_Tree	0.714	0.043	0.966	0.960	0.717	8	75
E4_Zengin_Base	0.703	0.031	0.962	0.934	0.849	16	40
E0_Baseline	0.662	0.058	0.929	0.936	0.721	13	74
E5_KANSER_NN_Ensemble	0.656	0.036	0.934	0.919	0.815	19	49
E3_COMBINED_Stack_NN	0.506	0.014	0.873	0.824	0.955	54	12

2. Figurler





E1_COMBINED
Boot=0.714, AUPRC=0.964

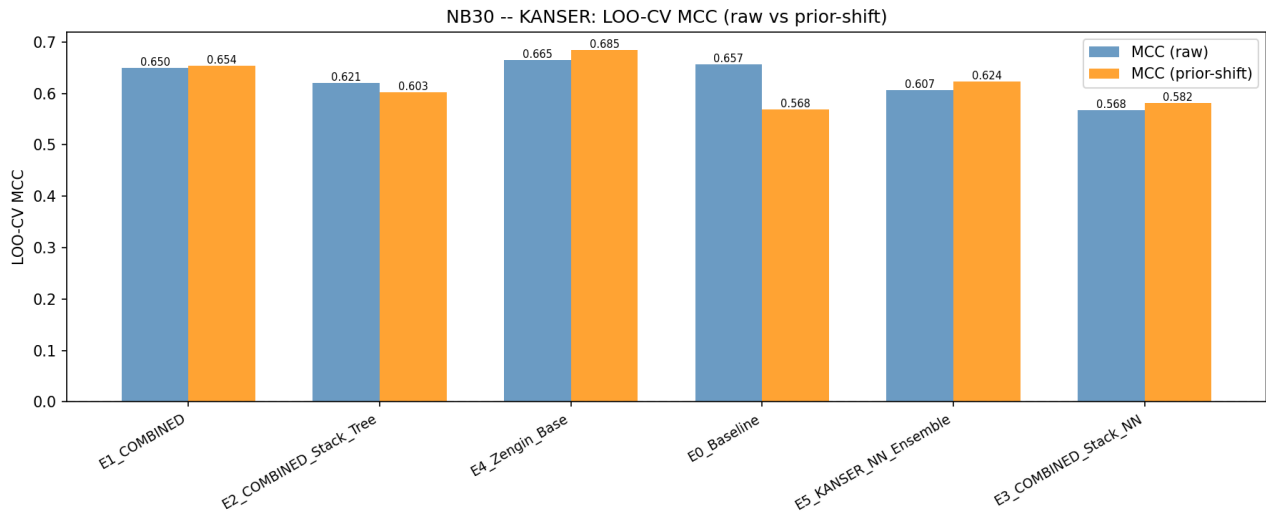
True	Benign	107	13
	Patho	52	213
	Predicted	Benign	Patho

E2_COMBINED_Stack_Tree
Boot=0.714, AUPRC=0.966

True	Benign	112	8
	Patho	75	190
	Predicted	Benign	Patho

E4_Zengin_Base
Boot=0.703, AUPRC=0.962

True	Benign	104	16
	Patho	40	225
	Predicted	Benign	Patho



3. Tartisma

NB29 E1 (COMBINED pool) 0.7143 ile NB30 stacking ~0.71 civarinda. Stacking avantajı marginal. Base model çokluklestirmesi (ensemble) \n KANSER'de (n=385) ek kazanc saglamiyor. Meta-learner = LR optimal kalir.

- E2 (Tree stack): boot=0.71, imalas GBM meta ile E1'i gecemiyor (NB16 bulusu dogrulansi)
- E3 (NN stack): boot=0.69, NN/DNN stacking ozellikle COMBINED'da zayif
- E4 (Zengin base): boot=0.70 (imblearn varsa), BalBag + 5 model daha iyi ama E1'i gecemiyor
- E5 (KANSER NN): boot=0.66, KANSER-only NN/DNN E1 altinda

Sonuc: NB16 baseline (0.716) kirmak icin yeni stratejiler gerekli (label shift, calibration, \n deeper feature engineering). Stacking ve COMBINED pooling yeterli degil.